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Title

INNOVATING CARDIOPULMONARY RESUSCITATION EDUCATION: ENHANCING PRACTICAL SKILLS WITH VIRTUAL REALITY.

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Abstract

Background: Virtual Reality (VR) uses 3D graphics and interactive devices to immerse users in virtual environments, enabling "safe fails" crucial for student learning and patient safety. Its popularity aligns with digital-native students' preferences. The Ministry of Health mandates CPR competency for all healthcare personnel based on international guidelines. Conventional CPR instruction with task trainer manikins focuses on technical skills but fails to represent realistic emergencies, leaving trainees unprepared. VR can simulate high-stress conditions, but existing VR CPR training software lacks technical precision. We created a combination of VR immersive simulations and task trainers' haptic feedback, offering unlimited practice and enhancing comprehension and retention in realistic settings. Based on Kolb's constructivist and experiential learning theories, it encourages active exploration, reflection, conceptualisation, and practical experience. Method: The VR-CPR project was conceptualised based on the CIPP framework, ensuring a structured approach to assessing, implementing and evaluating educational programmes. The VR-CPR application includes an introduction to VR training, a narrated tutorial on fundamental interactions, a step-by-step animated CPR module, and a realistic VR environment for situational analysis and feedback. The setup includes an HTC-Vive headset, motion sensors, and Vive Tracker 3.0 handsets for gesture recognition and control. The VR-CPR programme was integrated into the curriculum. Due to limited VR-headsets, students use them after theory and manikin-based skill teaching and before competency assessments. Online booking system allowed flexible, self-directed learning, maximising hands-on practice opportunities outside regular classroom hours. Feedback was gathered using a validated user experience questionnaire from 38 Semester 1 to 3 medical students. It measured Adaptation/Immersion, Involvement, Interface Quality, and Sensory Fidelity. Results showed generally positive experiences, with mean scores of 3.6 to 4.5 on a 6-point scale. Students reported good engagement, effective visual surveying, and minimal visual distractions but noted room for improvement in physical interaction and control device adjustment. Results: Focus group discussions were conducted with two groups of six Semester 5 students and one group of six faculty members. Results showed students found VR-CPR beneficial for understanding

and retention of CPR concepts when used alongside conventional methods. Despite initial technical issues, VR's interactive and engaging nature was appreciated. Both students and faculty agreed that VR should complement, not replace, conventional training. Faculty supported integrating VR-CPR but noted challenges with technology and suggested additional faculty training. Conclusion: Using the CIPP framework, VR-CPR integrates immersive simulations with tactile feedback to enhance learning outcomes. Feedback is positive, but challenges like technical issues and faculty training need to remain. Future research should compare VR with conventional methods and address long-term retention.

Subjects

[KOLB'S Experiential Learning theory](#); [CARDIOPULMONARY resuscitation education](#); [AUTODIDACTICISM](#); [RESERVATION systems](#); [MEDICAL personnel](#); [HAPTIC devices](#)

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